

MAIE5212 ASSIGNMENT 1

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Q1: Representation using perceptrons

Solution

Represent the boolean expression $\neg a \vee b \vee \neg c$ using a single perceptron.

Perceptron parameters:

◦ **Weights:**

$$w_a = -1 \quad (\text{for } a)$$

$$w_b = 1 \quad (\text{for } b)$$

$$w_c = -1 \quad (\text{for } c)$$

◦ **Bias term:** $b = 1$

◦ **Activation function:** Step function: $\text{step}(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$

The perceptron computes: $z = (-1) \cdot a + 1 \cdot b + (-1) \cdot c + 1$

Truth Table Verification

The perceptron correctly implements $\neg a \vee b \vee \neg c$,
as it outputs 0 only when $a = 1, b = 0, c = 1$.

a	b	c	Calculation	Output	Expected
0	0	0	$(-1) \times 0 + 1 \times 0 + (-1) \times 0 + 1 = 1$	1	1
0	0	1	$(-1) \times 0 + 1 \times 0 + (-1) \times 1 + 1 = 0$	1	1
0	1	0	$(-1) \times 0 + 1 \times 1 + (-1) \times 0 + 1 = 2$	1	1
0	1	1	$(-1) \times 0 + 1 \times 1 + (-1) \times 1 + 1 = 1$	1	1
1	0	0	$(-1) \times 1 + 1 \times 0 + (-1) \times 0 + 1 = 0$	1	1
1	0	1	$(-1) \times 1 + 1 \times 0 + (-1) \times 1 + 1 = -1$	0	0
1	1	0	$(-1) \times 1 + 1 \times 1 + (-1) \times 0 + 1 = 1$	1	1
1	1	1	$(-1) \times 1 + 1 \times 1 + (-1) \times 1 + 1 = 0$	1	1

Table 1: Truth table verification for $\neg a \vee b \vee \neg c$

Q2: Computation graph

Computation Graph

$$f = \text{sigmoid}(\max(x, y) + z \cdot w)$$

$$\text{sigmoid}(u) = \frac{1}{1 + e^{-u}}$$

Intermediate variables:

$$a = \max(x, y)$$

$$b = z \cdot w$$

$$c = a + b$$

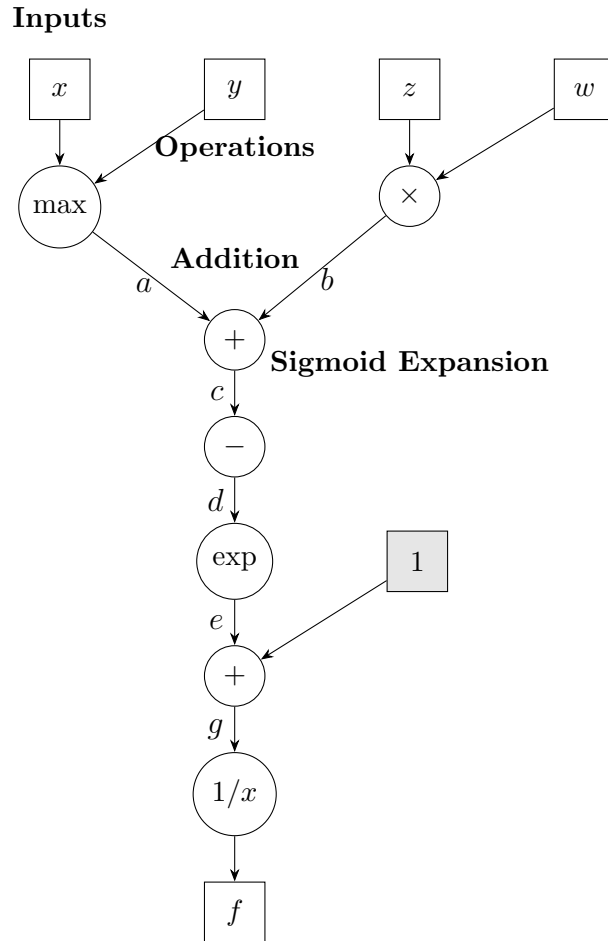
$$d = -c$$

$$e = \exp(d)$$

$$g = 1 + e$$

$$f = 1/g$$

The computation graph with intermediate variables:



Forward Pass

Forward Pass with $x = 1, y = -2, z = 0.5, w = 2$:

$$a = \max(1, -2) = 1$$

$$b = 0.5 \times 2 = 1$$

$$c = 1 + 1 = 2$$

$$d = -2$$

$$e = e^{-2} \approx 0.135335$$

$$g = 1 + 0.135335 = 1.135335$$

$$f = 1/1.135335 \approx 0.880797$$

Backward Pass

$$\text{for sigmoid } \frac{\partial f}{\partial c} = f(1 - f)$$

$$\frac{\partial f}{\partial c} = f(1 - f) \approx 0.880797 \times (1 - 0.880797) \approx 0.104994$$

$$\frac{\partial c}{\partial a} = 1, \quad \frac{\partial c}{\partial b} = 1$$

$$\frac{\partial a}{\partial x} = \begin{cases} 1 & \text{if } x \geq y \\ 0 & \text{otherwise} \end{cases} = 1 \quad (\text{since } 1 > -2)$$

$$\frac{\partial a}{\partial y} = \begin{cases} 0 & \text{if } x \geq y \\ 1 & \text{otherwise} \end{cases} = 0 \quad (\text{since } 1 > -2)$$

$$\frac{\partial b}{\partial z} = w = 2, \quad \frac{\partial b}{\partial w} = z = 0.5$$

Compute the final gradients:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial a} \cdot \frac{\partial a}{\partial x} = 0.104994 \times 1 \times 1 \approx 0.104994$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial a} \cdot \frac{\partial a}{\partial y} = 0.104994 \times 1 \times 0 = 0$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial b} \cdot \frac{\partial b}{\partial z} = 0.104994 \times 1 \times 2 \approx 0.209988$$

$$\frac{\partial f}{\partial w} = \frac{\partial f}{\partial c} \cdot \frac{\partial c}{\partial b} \cdot \frac{\partial b}{\partial w} = 0.104994 \times 1 \times 0.5 \approx 0.052497$$